Thomas Qi

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Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science — *09/2023 – 04/2028*
- **GPA:** 3.92/4.0 (91.1%)
- **Coursework:** Operating Systems, Concurrency, Databases, Algorithms, Data Structures, OOP, AI, ML, Computer Architecture

Experience

Point72 — New York, NY
Quantitative Developer Intern — *01/2027 – 04/2027*
- Incoming winter 2027, Internal Alpha Capture (systematic long/short equity).

Coinbase — San Francisco, CA
Software Engineer Intern — *05/2026 – 08/2026*
- Built an AI oncall copilot using **LangGraph**, FastAPI, and a **Redis**-backed job queue to automatically triage PagerDuty alerts via Slack, integrating Datadog and MESDb with RAG-based context retrieval; used by 150+ engineers to reduce MTTR by 43%.
- Designed a self-improving **RAG** pipeline with a **pgvector** knowledge base that ingests engineer feedback via Slack reactions, embedding validated resolutions and pruning stale context to continuously improve triage accuracy across 200+ incident types.

BitGo — Palo Alto, CA
Software Engineer Intern – Electronic Trading — *09/2025 – 12/2025*
- Engineered a cron-scheduled **Go** ETL pipeline that extracts, normalizes, & reconciles 2M+ user trades into **PostgreSQL**, enabling automatic LP settlement & eliminating a manual workflow to save ops 5 hours/week.
- Built an internal **NextJS** & **GraphQL** dashboard to visualize 2M+ trades, enabling ops to resolve anomalies 2.8× faster; cutting median load time by 58% (384ms to 163ms) using server-side pagination.
- Integrated **Prometheus** counters + SLO alerts for 3 services, cutting MTTR by 53% & alert noise by 66%.

University of Waterloo — Waterloo, ON
Research Intern – Coding Agent Tooling — *05/2025 – 12/2025*
- Developed a **Python Selenium** web scraper that gathered issue and commit data from 350+ GitHub repositories (75K+ commits) to construct benchmarks evaluating CodeLLMs on real refactors and bug-fixes.
- Developed a language-agnostic parser & MCP Server that improves AI coding agent problem-solving benchmarks by 5% (on average), by enabling LLMs to query call-graphs for large repositories.

Habtech Communications — Mississauga, ON
Software Engineer Intern — *01/2025 – 04/2025*
- Developed a construction project estimation system serving 250+ engineers & estimators across 3 companies, generating **$185,000+** in licensing revenue; built with **NextJS**, **ASP.NET**, **MS SQL Server**.
- Deployed a **Jenkins** server on EC2 to enable CI/CD pipelines for 4 applications on **Docker**, automating container building & deployment from Git to Azure to save 5+ hours of monthly manual build time.

Atomic Dev Inc. — Los Angeles, CA
Software Engineer Intern — *04/2024 – 08/2024*
- Engineered & deployed an internal tool that accurately generates SQL migration scripts from git logs, saving 5+ hours of weekly labour using chain-of-thought prompting, **NextJS**, **OpenAI API**, **AWS EC2**.

Projects

SyncSpark | ***(NextJS, Express, Socket.io, Redux, TailwindCSS, Docker, AWS)***
- Engineered a full-stack, low-latency real-time collaborative code editor that lets multiple developers edit & run code together & chat live using Monaco Editor + **Websockets** (< 50 ms average round-trip).

Skills

Languages: C/C++, Python, Go, TypeScript, C#, SQL, GraphQL
Libraries/Frameworks: ASP.NET, NextJS, Express, Redux, TailwindCSS, Websockets, Entity Framework, Flask
Developer Tools: AWS, Redis, Kafka, Kubernetes, Docker, Git, SQL Server, PostgreSQL, Azure, Vercel